

Decoupling the Feedback and Feedforward Components of Impedance Controllers: Theory and Experiments in Lower-Limb Prostheses

T. Kevin Best, Elliott J Rouse, Robert D Gregg

Abstract

Robotic prostheses commonly use joint-space impedance controllers parameterized by stiffness, damping, and an equilibrium angle to create desired behaviors. Although these controllers are often interpreted as equilibrium angle tracking controllers, their parameters are chosen such that ground and user interactions cause a different kinematic pattern to emerge, complicating their design, tuning, and interpretation. To address this challenge, we introduce an alternative formulation of the impedance controller comprising both a feedback position control term and feedforward torque control term. This equivalent form clarifies how the impedance parameters shape both the nominal and perturbed behaviors of the controller. In both theory and experiments with an above-knee amputee participant, we demonstrate that controllers with appropriately designed feedforward torque components can produce identical nominal behaviors despite differences in stiffness and damping, which primarily govern how the system responds to perturbations. Our findings offer important insights for prosthesis controller design and tuning: 1) our decoupled parameterization allows independent prescription of an impedance controller's nominal and off-nominal behaviors; 2) tuning stiffness and damping based on nominal walking alone is insufficient; and 3) even non-impedance paradigms can benefit from applying impedance concepts to achieve robust real-world behavior.

Keywords

impedance control, prostheses, robotics and control systems

Decoupling the Feedback and Feedforward Components of Impedance Controllers: Theory and Experiments in Lower-Limb Prostheses

T. Kevin Best, Elliott J. Rouse*, and Robert D. Gregg*

Abstract—Robotic prostheses commonly use joint-space impedance controllers parameterized by stiffness, damping, and an equilibrium angle to create desired behaviors. Although these controllers are often interpreted as equilibrium angle tracking controllers, their parameters are chosen such that ground and user interactions cause a different kinematic pattern to emerge, complicating their design, tuning, and interpretation. To address this challenge, we introduce an alternative formulation of the impedance controller comprising both a feedback position control term and feedforward torque control term. This equivalent form clarifies how the impedance parameters shape both the nominal and perturbed behaviors of the controller. In both theory and experiments with an above-knee amputee participant, we demonstrate that controllers with appropriately designed feedforward torque components can produce identical nominal behaviors despite differences in stiffness and damping, which primarily govern how the system responds to perturbations. Our findings offer important insights for prosthesis controller design and tuning: 1) our decoupled parameterization allows independent prescription of an impedance controller’s nominal and off-nominal behaviors; 2) tuning stiffness and damping based on nominal walking alone is insufficient; and 3) even non-impedance paradigms can benefit from applying impedance concepts to achieve robust real-world behavior.

I. INTRODUCTION

ROBOTIC lower-limb prostheses are a promising emerging technology with the potential to significantly improve mobility and quality of life for millions of people living with a lower-limb amputation by better replacing the functions of the missing limb and its musculature. A critical component in their success is the control instructions—or *policy*—that encodes how and when to apply actuator torques to create the desired locomotion patterns. Impedance control is commonly used for this purpose, and it has seen widespread usage in both laboratory [1], [2] and commercial [3] applications.

Traditionally, the joint-space impedance controllers¹ used in

robotic lower-limb prostheses have had the form of

$$\tau = -K(\theta - \theta_{eq}) - B\dot{\theta}, \quad (1)$$

where the joint torque commands τ use feedback from the joint angle θ and velocity $\dot{\theta}$ to emulate the dynamics of time-varying rotary spring-damper systems with stiffness K , damping B , and a rest position or equilibrium angle θ_{eq} . The values of these *impedance parameters* change throughout the gait cycle to produce the desired locomotion behaviors. Their evolution is often determined by a finite state machine (e.g., [5]–[7]) or a continuous phase representation (e.g., [8]–[10]). For conciseness, we omit the explicit functional dependence in the notation for the impedance parameters, although they are generally understood to vary over time (e.g., $K = K(t)$ or $K = K(\phi(t))$ where $\phi(t)$ is a phase variable [11], [12]).

At first glance, (1) appears to be a simple joint-state feedback controller that drives $\theta \rightarrow \theta_{eq}$, similar to a mechanical spring-damper system. In fact, that is indeed what would happen if the joint were allowed to move in isolation without external interactions from the ground or the user. In practice however, the prosthesis’s joints are part of a larger dynamical system comprising the prosthesis, the user, and the environment. These other system components create interaction torques τ_{ext} that have a substantial effect on the joint’s movement patterns, causing the actual joint trajectory $\theta(t)$ to often differ from $\theta_{eq}(t)$. Therefore, a key distinction in prosthesis impedance control is that the equilibrium trajectory $\theta_{eq}(t)$ is often designed to be distinct from the true, desired joint trajectory $\theta_d(t)$. That is, $\theta_{eq}(t)$ is not a target trajectory; it is not meant to describe the desired kinematic configuration of the limb. Rather, $\theta_{eq}(t)$ should be interpreted as a control input that, in conjunction with the other impedance parameters and the expected external interaction torques, results in the prosthesis’s joints following the desired trajectories. Recognizing this subtle yet fundamental distinction between θ_{eq} and θ_d is central to understanding how to design and tune prosthesis impedance controller behaviors.

However, developing such an understanding using the standard parameterization of the impedance controller given in (1) can be difficult. First, θ_d does not explicitly appear in (1), so it is unclear from simply looking at K , B , and θ_{eq} what kinematic patterns the controller was designed to produce. A similar ambiguity extends to the designed joint kinetics. Further, when designing an impedance controller for a specific behavior, one cannot simply select K , B , and θ_{eq} by observing a desired motion. While suitable values can be calculated from demonstration data (e.g., [8], [9], [13], [14]), these impedance parameters are not directly observable in the same way that joint kinematics and kinetics are observable through laboratory measurements. In many cases, suitable impedance parameters

This material is based upon work supported by the National Science Foundation under the Graduate Research Fellowship DGE 1841052, CAREER Award 1846969, and NRI Award Number 2024237. This material was also supported by the National Institute of Child Health & Human Development of the NIH under Award R01HD094772.

The authors are with the Department of Robotics at the University of Michigan, Ann Arbor, MI 48109. Elliott Rouse and Robert Gregg are also with the Department of Mechanical Engineering at the University of Michigan, Ann Arbor, MI 48019. Contact: tkbest@umich.edu, ejrouse@umich.edu, rdgregg@umich.edu.

*These authors contributed equally to this work.

¹Some readers may be familiar with impedance controllers in the context of robotic manipulators, like those proposed by Hogan in [4]. These controllers typically aim to emulate specified dynamics at the robot’s end-effector in a world-based coordinate system termed the *task-space*. The impedance controllers typically used in robotic lower-limb prostheses differ by operating in the *joint-space*, where the torque command for each joint is defined using only collocated kinematic feedback rather than through a Jacobian mapping.

are identified using iterative, empirical tuning rather than explicit design. This process is complicated by the fact that K , B , and θ_{eq} are non-unique for a given nominal motion, as we will show in this article. In total, these limitations make understanding, designing, and tuning impedance controllers with the traditional parameterization unnecessarily difficult.

To address these limitations, this article introduces a new formulation of the impedance controller that decouples its feedback and feedforward properties, making it better suited for design, tuning, and interpretation. Using this novel parameterization, we:

- 1) Show that impedance controllers are a composition of a feedback position controller and a feedforward torque controller, contextualizing them between impedance and torque control paradigms.
- 2) Derive and formalize key properties that clarify the distinct roles of the impedance parameters.
- 3) Demonstrate, both theoretically and experimentally, that different impedance parameters can yield identical nominal behaviors but markedly different responses to gait perturbations.
- 4) Interpret these findings in the context of impedance controller design, evaluation, and tuning for robust real-world use.

The article is organized as follows: Section II presents the mathematical derivations and the key theoretical results. Section III then presents an experimental validation of these results with an above-knee prosthesis user. Finally, Section IV explores the implications of our findings for prosthesis controller design and tuning.

II. THEORETICAL DEVELOPMENT

A. Impedance as Composite Torque and Position Control

As explained previously, an impedance controller's equilibrium angle θ_{eq} is not necessarily the same as the desired joint angle θ_d in typical robotic prosthesis control. Let us define the difference between the equilibrium angle and the desired angle, which we will term the *deviation angle* $\hat{\theta}$, as

$$\hat{\theta} := \theta_{eq} - \theta_d. \quad (2)$$

Substituting in this definition, adding and subtracting $B\dot{\theta}_d$, and then grouping terms, we can rewrite the traditional impedance control law (1) as the sum of a standard proportional-derivative (PD) position controller and a feedforward torque controller that is independent of the present kinematics:

$$\begin{aligned} \tau &= -K(\theta - \hat{\theta} - \theta_d) - B\dot{\theta} + B\dot{\theta}_d - B\dot{\theta}_d \\ &= \underbrace{-K(\theta - \theta_d) - B(\dot{\theta} - \dot{\theta}_d)}_{\text{feedback}} + \underbrace{K\hat{\theta} - B\dot{\theta}_d}_{\text{feedforward}}. \end{aligned}$$

As $\hat{\theta}$ is a free parameter, the term $K\hat{\theta} - B\dot{\theta}_d$ can be arbitrarily selected regardless of K and B . Let us define this quantity as a new independent parameter $\tau_{ff} := K\hat{\theta} - B\dot{\theta}_d$, resulting in the new impedance control law:

$$\tau = -K(\theta - \theta_d) - B(\dot{\theta} - \dot{\theta}_d) + \tau_{ff}. \quad (3)$$

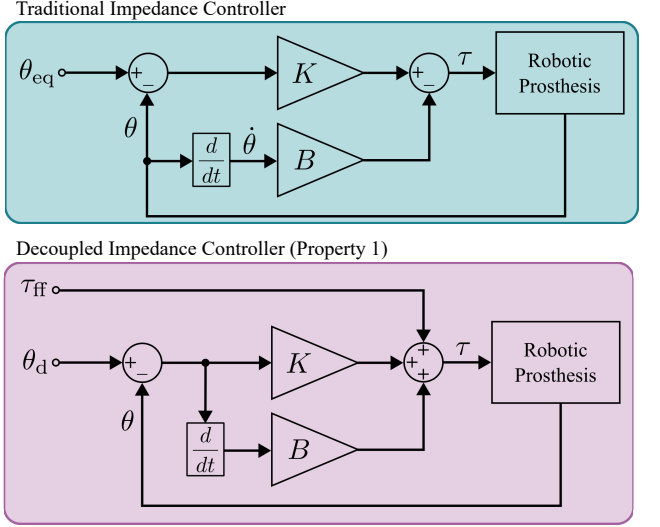


Fig. 1. Block diagrams comparing the two mathematically equivalent interpretations of the impedance controller. (Top) The traditional impedance control law (1) regulates the measured joint angle θ to the equilibrium angle θ_{eq} with a stiffness K . All joint velocity is damped with a viscosity B . (Bottom) An alternative, equivalent interpretation of the impedance controller (3) proposed in Property 1 where the joint angle and velocity are regulated to a desired kinematic trajectory θ_d using a standard PD position controller with a proportional gain K and derivative gain B . A feedforward torque τ_{ff} is added to the output of the feedback position controller.

From this perspective, an impedance controller is nothing more than the sum of a feedback PD position controller regulating the joint angle around a desired trajectory and a feedforward torque controller with an output that is independent of the motion of the joint (Fig. 1). For future use, let us explicitly denote this fact as Property 1:

Property 1. *An impedance controller is mathematically equivalent to a feedforward torque controller combined with a proportional derivative position controller.*

This new impedance control law (3) is parameterized by K , B , τ_{ff} , and a nominal desired trajectory $\theta_d, \dot{\theta}_d$. As with the standard form, these quantities may in general be arbitrary functions of time, although we omit the explicit time dependence in our notation for brevity (i.e. $\tau_{ff} = \tau_{ff}(t)$). To transform a controller parameterized as (3) back to (1), the equilibrium angle can be recovered from these parameters as

$$\theta_{eq} = \theta_d + (\tau_{ff} + B\dot{\theta}_d)/K. \quad (4)$$

B. The Relationships Between Torque, Impedance, and Position Control Paradigms

Property 1 explicitly shows that the key distinction between an impedance controller and a PD position controller is the presence of the feedforward torque term τ_{ff} . When $\tau_{ff} \neq 0$, an impedance controller can generate non-zero torque commands even while tracking the desired kinematics. This enables the controller to proactively account for expected dynamics (e.g., gravitational or inertial torques) and interaction torques (e.g., those imposed by the ground and user) without requiring deviations from the desired kinematics. In contrast, a PD controller can only produce output torque in response to observed

kinematic errors, meaning that tracking error is *required* in the presence of dynamics and interaction torques.² This purely reactive nature of PD controllers can also lead to inevitable phase lag between the desired time-varying trajectory and the actual joint positions. Viewed from this perspective, a PD position controller is simply a special case of the impedance controller where $\tau_{ff} = 0$.

Less immediately clear from (3) is how impedance controllers relate to more general torque controllers. Consider an arbitrary torque control policy $\tau = f(\theta, \dot{\theta}, \xi)$, where ξ contains any number of additional inputs (*e.g.*, gait phase, electromyography measurements, time, etc.). Taking the Taylor series expansion about a nominal trajectory $\theta_d, \dot{\theta}_d, \xi_d$ with respect to the joint kinematics, we find

$$\tau = f(\theta_d, \dot{\theta}_d, \xi_d) + \frac{\partial f}{\partial \theta} (\theta - \theta_d) + \frac{\partial f}{\partial \dot{\theta}} (\dot{\theta} - \dot{\theta}_d) + \text{H.O.T.} \quad (5)$$

If we neglect the higher-order terms (H.O.T.), we can equate the coefficients between (3) and (5):

$$\tau_{ff} = f(\theta_d, \dot{\theta}_d, \xi_d), \quad K = -\frac{\partial f}{\partial \theta}, \quad B = -\frac{\partial f}{\partial \dot{\theta}}. \quad (6)$$

Thus, an impedance controller can provide a first-order approximation to *any* torque control policy about a nominal trajectory. The stiffness and damping parameters are simply the first partial derivatives of f with respect to the kinematics. To find the impedance trajectories, one only needs to evaluate these functions along the nominal trajectories. Let us define this result as Property 2:

Property 2. *An impedance controller can provide a first-order approximation to any arbitrary control policy $\tau = f(\theta, \dot{\theta}, \xi)$ with respect to position and velocity about a nominal trajectory $\theta_d, \dot{\theta}_d, \xi_d$.*

To illustrate this property, consider a neural network control policy $\tau = \pi_1(\theta_a, \dot{\theta}_a, \theta_f, \dot{\theta}_f)$ that is trained to output biomimetic ankle torque commands based on measurements of the relative ankle angle θ_a and the global foot angle θ_f , as well as their respective time derivatives $\dot{\theta}_a$ and $\dot{\theta}_f$. The example network in Fig. 2a has four hidden layers of sizes 100, 50, 10, and 5, and it uses ReLU activation. We trained the network based on the able-bodied variable ramp and speed walking data reported in [15]. The resulting control policy is highly nonlinear and analytically appears nothing like the standard impedance controller (1).

For a specific expected kinematic trajectory, such as the average level ground 1.0 m/s walking data from [15], we can calculate the policy's K and B trajectories using (6) (Fig. 2b).³ One could also calculate θ_{eq} via (4), but since there

are no guarantees that $K \neq 0$, it is numerically safer to use the form given in (3) and calculate τ_{ff} from (6). We can use these impedance trajectories to define an impedance control law π_2 , which locally approximates π_1 in the neighborhood of the expected kinematics (Fig. 2c). When provided with all level ground 1.0 m/s walking strides in the dataset, we see that both the neural network and the impedance controller produce similar average torque responses (Fig. 2d). However, π_2 does not capture all of the variance observed in π_1 , as it is simply a local (linear) approximation.

If we apply a momentary 5 deg step dorsiflexion perturbation during midstance to the input ankle angle data, we see that both control policies respond similarly with a corresponding increase in ankle torque (Fig. 2e). However, π_2 only approximates π_1 to the first order, so we see minor differences between the two torque responses near the peak plantarflexion torque. These differences occur because π_2 neglects the higher order terms and feedback on θ_f and $\dot{\theta}_f$.

C. The Roles of the Impedance Parameters

With this theoretical foundation established, we can now address an important question that is fundamental to the interpretation, design, and tuning of impedance controllers: how does changing the different impedance parameters affect the controller's behavior? In the traditional spring-damper parameterization (1), this question is difficult to answer because the feedback and feedforward behaviors are coupled together, making the specific roles of each parameter unclear. To illustrate this limitation, it is helpful to consider two distinct possible operating conditions: nominal behavior (*i.e.* when $\theta = \theta_d$ and $\dot{\theta} = \dot{\theta}_d$) and off-nominal behavior (*i.e.* when $\theta \neq \theta_d$ or $\dot{\theta} \neq \dot{\theta}_d$). The controller's behavior in each case can be directly determined from the parameters of the decoupled parameterization (3) (*i.e.* $\{K, B, \tau_{ff}, \theta_d, \dot{\theta}_d\}$), which we explore below.

1) *Nominal Conditions:* During nominal conditions, we can see that the feedback component of the impedance controller in (3) goes to zero, and the torque command reduces to simply $\tau = \tau_{ff}$. Thus, we can interpret τ_{ff} as the torque that the impedance controller will produce when the desired kinematics are tracked (when $\theta = \theta_d$ and $\dot{\theta} = \dot{\theta}_d$). Since the feedback term becomes zero during nominal conditions regardless of the values of K and B , it means that their values can be varied arbitrarily without affecting the resulting torque command. Let us formalize this fact as Property 3:

Property 3. *When $\theta = \theta_d$ and $\dot{\theta} = \dot{\theta}_d$, the impedance controller reduces to $\tau = \tau_{ff}$, and the stiffness and damping parameters can be arbitrarily varied without changing the resulting controller output.*

A potentially surprising result follows from this property. Let us assume that we know the correct torque command $\tau^*(t)$ to apply in order to make the human-prosthesis system evolve along the desired kinematics $\theta_d(t)$, $\dot{\theta}_d(t)$. Datasets of able-bodied locomotion provide ample examples of joint torque trajectories that produce various locomotion behaviors

²Often, an integral term is added to a PD controller when a non-zero control output is required to maintain the desired reference. However, this approach works best when the required output is constant, as the only way to change the integral term's output is to encounter tracking errors. Alternatively, feedforward control inputs can preemptively change their torque contribution without requiring tracking error, which works well in cyclic and predictable applications. The downside, of course, is that they cannot modify their behavior when conditions are different than expected.

³The presence of negative stiffness and damping values will be discussed in Section IV.

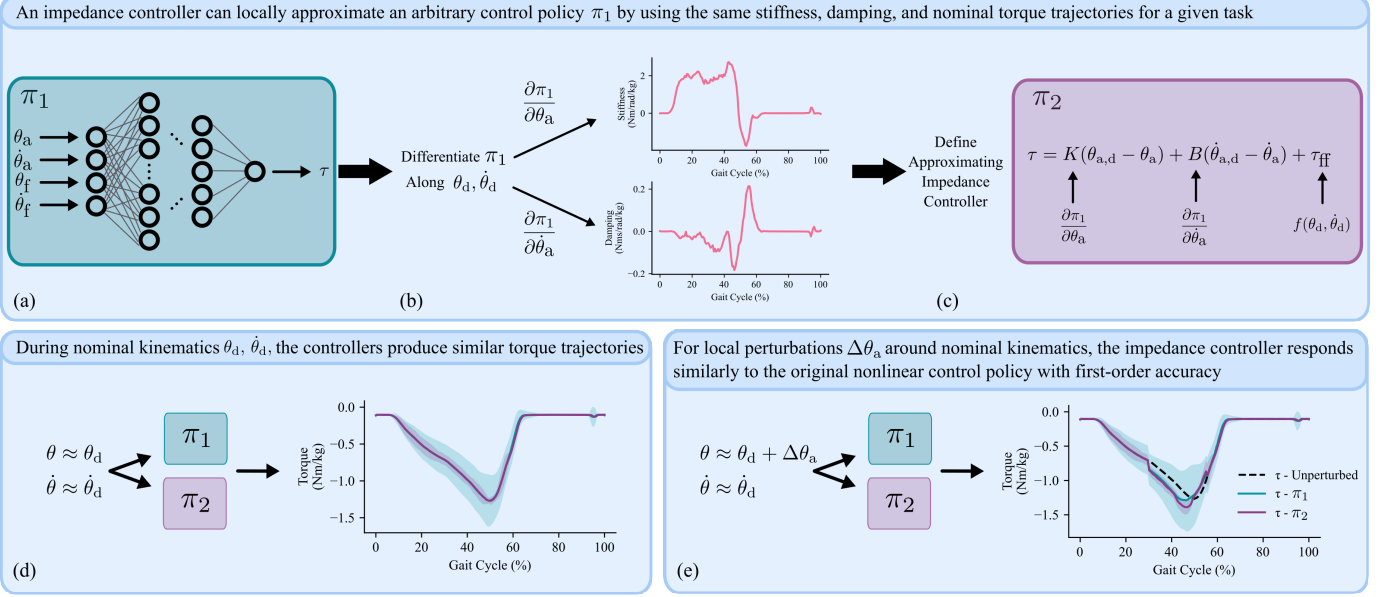


Fig. 2. An example illustrating how impedance controllers can provide a first-order approximation to an arbitrary nonlinear torque control policy about a nominal trajectory. (a) A neural network control policy π_1 maps foot and ankle kinematic inputs to ankle torque, which was trained on variable ramp and speed walking data from [15]. (b) The average stiffness and damping trajectories of the neural network control policy evaluated along a specific nominal task trajectory using (6). In this example, we use the specific task of 1.0 m/s level-ground walking. (c) An impedance controller π_2 that locally approximates π_1 for this task. (d) The average torque output from both control policies matches when provided with all 1.0 m/s level-ground walking strides from the dataset. (e) Both policies respond similarly to a momentary 5.0 deg dorsiflexion perturbation applied to the nominal input kinematics during midstance, but π_2 only approximates π_1 to first-order accuracy. Shaded regions represent ± 1 standard deviation.

[15], [16]. Alternatively, reference torque commands for a given kinematic pattern can be calculated using dynamic models and trajectory optimization. If we select $\tau_{ff}(t) = \tau^*(t)$ and we start with nominal initial conditions $\theta(0) = \theta_d(0)$ and $\dot{\theta}(0) = \dot{\theta}_d(0)$, (3) will reduce to simply $\tau = \tau^*(t)$ via Property 3, and we will, by assumption, observe that $\theta(t) = \theta_d(t), \dot{\theta}(t) = \dot{\theta}_d(t), \forall t \geq 0$. In other words, if the feedforward torque command is chosen correctly such that the desired kinematic trajectory is followed, the stiffness and damping values will have no effect on the output torque command, and their values can be varied arbitrarily without affecting the observed kinematic and kinetic behavior. Let us formalize this point in the following Corollary:

Corollary 3.1. *If two controllers share a common τ_{ff} trajectory that makes the system evolve along the desired kinematics $\theta_d, \dot{\theta}_d$, they will produce the same torque and angle behavior regardless of differences in stiffness and damping.*

This result highlights the important distinction between joint *stiffness* and *quasi-stiffness* [17]. The quasi-stiffness describes the slope of the curve traced in the torque-angle plane over time during nominal conditions. In contrast, the stiffness describes how the torque command instantaneously changes due to kinematic changes (i.e. $-\frac{\partial \tau}{\partial \theta}$). Controllers that have the same τ_{ff} producing the desired kinematics have identical quasi-stiffness, regardless of any differences in stiffness K .

To demonstrate Corollary 3.1, we designed two impedance controllers for a robotic ankle prosthesis that have very different stiffness and damping trajectories. Controller α used the biological ankle stiffness and damping trajectories reported in [18] while controller β instead used constant stiffness and

damping values of medium magnitudes (Fig. 3a, top/middle). For both controllers, we set $\theta_d, \dot{\theta}_d$ and τ_{ff} to match the nominal able-bodied ankle kinematic and kinetic behavior for level ground 1.0 m/s walking reported in [15]. Using (4), we can calculate θ_{eq} from these parameters, showing that it is also quite different between controllers (Fig. 3a, bottom). If we assume that applying τ_{ff} will produce the nominal kinematics, as the dataset suggests, we can see that both controllers produce identical torque and angle behaviors despite very different values for K , B , and θ_{eq} (Fig. 3b).

2) *Off-Nominal Conditions:* During off-nominal conditions (i.e. when $\theta \neq \theta_d$ or $\dot{\theta} \neq \dot{\theta}_d$), the stiffness K and damping B parameters remain in (3) and affect the output torque command. While they do not affect the controller's behavior during nominal conditions, these parameters determine how the torque command changes when kinematic deviations occur. One can think of the feedback component as a stabilizing controller around the nominal trajectory, with K and B determining the dynamics of that stabilization. If we define the relative “impedance magnitude” of the controller as $\|Z\| = |K| + |B|$, we see that impedance controllers can span a continuum between low and high impedance behaviors (Fig. 4). If K and B are large, the controller's feedback component will respond aggressively to any deviation from the nominal kinematics. Thus, the controller will behave similarly to a typical high-gain PD position controller, aside from the presence of τ_{ff} . On the other extreme, low values of K and B will result in low-gain feedback components that yield little effect in response to kinematic variation, resulting in a controller that is overall driven by the feedforward commands τ_{ff} . As the choice of K and B is independent from the desired nominal behavior

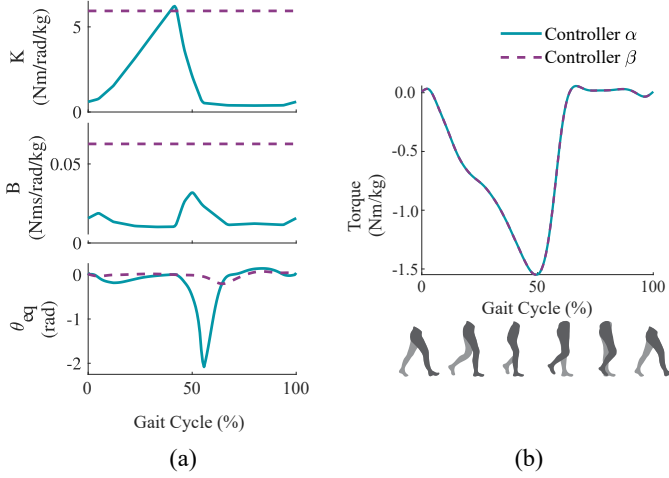


Fig. 3. (a) The impedance parameter trajectories of two different ankle impedance controllers (α & β), both designed for level ground 1.0 m/s walking. (b) The identical torque responses of the two controllers when subjected to nominal walking kinematics, despite their different impedance parameters.

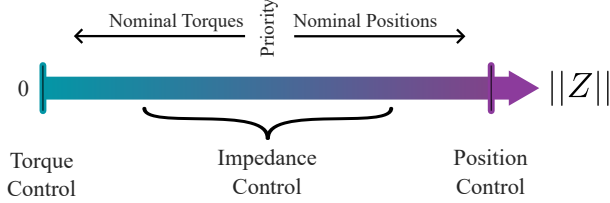


Fig. 4. A conceptual diagram depicting how torque control, impedance control, and position control are related along the mechanical impedance magnitude spectrum, $||Z||$. The impedance parameters determine the controller's $||Z||$, scaling the controller's behavior between open-loop torque control and high-feedback position control. Most "impedance controllers" in the literature lie somewhere in the middle of this spectrum. Control policies on the left side of the spectrum prioritize nominal joint torques whereas policies on the right side prioritize nominal joint positions.

(see Property 3), the controller's relative impedance can be considered as a design knob that trades between enforcing nominal joint torques and nominal joint positions.

This behavior becomes clear when examining how two different controllers respond to kinematic deviations. Returning to the ankle controller example from earlier, both control policies created identical torque-angle curves (*i.e.* quasi-stiffness) under nominal conditions despite their differing stiffness and damping. However, we can observe their impedance differences (*i.e.* K and B) by examining how their torque output changes in response to perturbations in position $\Delta\theta_a$ and velocity $\Delta\dot{\theta}_a$ across the gait cycle. Near heelstrike for example, Controller β responds more aggressively to changes in position $\Delta\theta_a$ than Controller α . This is because Controller β has a higher stiffness at this point in the gait cycle (Fig. 3a). In fact, each controller is a distinct surface (Fig. 5), where the slope of the surface in the torque and position (or velocity) perturbation plane is defined by the controller's stiffness (or damping). The two control surfaces intersect along the nominal behavior curve (*i.e.* when $\Delta\theta_a = 0$ and $\Delta\dot{\theta}_a = 0$), which is the same as Fig. 3b.

III. EXPERIMENTAL VALIDATION

To validate these properties and show how they apply in practice, we enrolled a participant with an above knee amputation to participate in walking experiments with the Open-Source Leg (OSL, second generation [19]). The experimental protocol was approved by the University of Michigan Institutional Review Board (HUM00143080). The participant (male, 25 years, 175 cm tall, 90 kg) provided informed consent and wore a safety harness during all trials to prevent falls.

A. Methods

We designed two walking controllers, α and β , which both shared common nominal behaviors (*i.e.* the same θ_d , $\dot{\theta}_d$, and τ_{ff}) but had different stiffness and damping trajectories. The trajectories for θ_d , $\dot{\theta}_d$, and τ_{ff} were taken as the average able-bodied values for 1.0 m/s level walking reported in [15], with the normalized kinetics scaled by our participant's mass. Like the examples above, controller α used the able-bodied stiffness and damping trajectories for the knee and ankle joints reported in [20] and [21]–[23], respectively (Fig. 6a). Controller β instead used constant stiffness and damping values of medium magnitudes. To synchronize the prosthesis's control behavior with the user's gait, we parameterized K , B , and θ_{eq} as functions of a phase variable ϕ , which converts the measured orientation of the user's residual thigh into an estimate of their gait cycle progression as in [9]. To ensure that the desired impedance was rendered accurately, we utilized the drivetrain dynamics compensation methods reported in [19].

Sagittal plane joint angles and moments for the OSL and biological joints were calculated from full-body motion capture and ground reaction force data. Marker trajectories and force plate signals were collected via a 26-camera motion capture system (Vicon, Oxford, UK) and low-pass filtered using a fourth-order zero-lag Butterworth filter at a 60 Hz cutoff frequency. Inverse kinematics (IK) and dynamics (ID) routines were performed in OpenSim 4.5 using a modified version of the native Rajagopal2016 musculoskeletal model adapted to incorporate the OSL v2 prosthesis's geometric and inertial properties, similar to [24], [25].

First, we asked the participant to walk at steady state on a treadmill at the designed speed of 1.0 m/s using each controller. Next, we induced kinematic deviations in the prosthesis's joints by quickly decelerating the prosthesis-side treadmill belt immediately after heelstrike during random strides. The belt-speed perturbations were rare in order to prevent anticipatory compensations from the participant. Perturbation magnitudes were set to $\pm 50\%$ of the measured steady-state walking speed and held for a duration corresponding to 25% of the total estimated gait cycle time.

B. Results

Despite the large differences in K , B , and θ_{eq} (Fig. 6a), the two controllers produced largely similar behaviors during nominal treadmill walking (Fig. 6b). This experimental result validates the theoretical result of Corollary 3.1. Some slight differences in the observed behaviors will be addressed in the

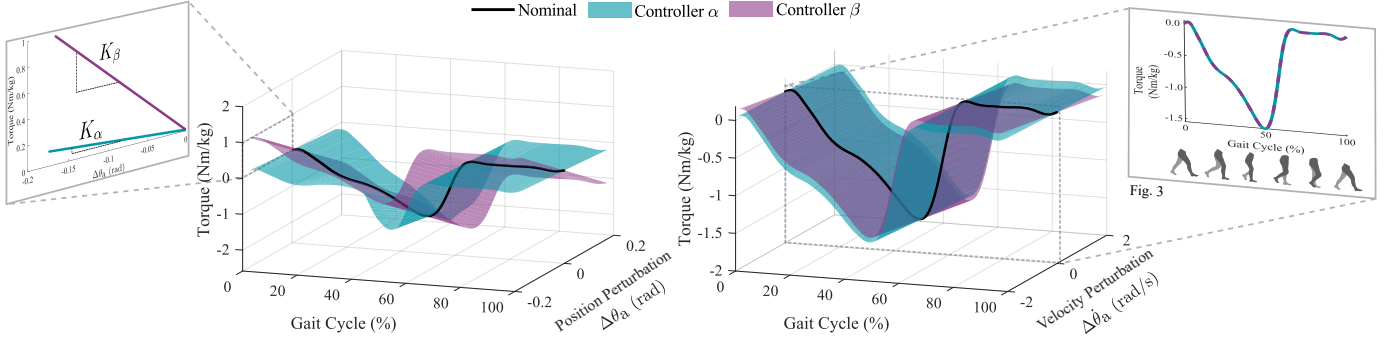


Fig. 5. Plots illustrating how two impedance controllers with different impedance parameters (controllers α and β from Fig. 3a) can produce the same nominal torque angle behavior (*i.e.* the same quasi-stiffness) but also different responses to angle $\Delta\theta_a$ (left) and velocity $\Delta\dot{\theta}_a$ (right) perturbations. Along the nominal conditions ($\Delta\theta_a = \Delta\dot{\theta}_a = 0$), the surfaces intersect and both controllers create the same output torque (Fig. 3b). Away from nominal conditions, the surfaces depart because they have different stiffness and damping trajectories. The slopes of the control surfaces in the torque/perturbation plane are defined by the controllers' stiffness (left) and damping (right).

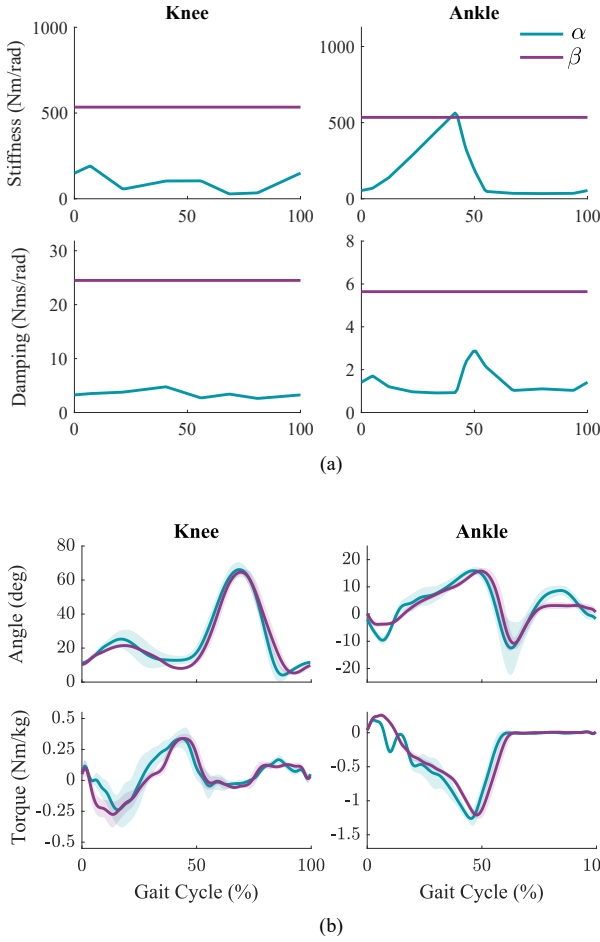


Fig. 6. (a) The stiffness and damping trajectories for each controller, α and β . (b) The average nominal kinematics and kinetics produced by each controller during level 1.0 m/s walking on a treadmill. Shaded regions represent ± 1 standard deviation.

Discussion section, but overall the two controllers produced nearly identical kinematics and kinetics.

During the perturbed strides, however, the kinematic and kinetic behavior of the prosthesis differed substantially between the two control conditions (Fig. 7), again consistent with our theoretical results. Placing both controllers along

the impedance spectrum (Fig. 4), we see that controller α , with its low impedance magnitude in early stance, is more like a feedforward torque controller. In contrast, controller β , with higher stiffness and damping, is closer to the behavior of a position controller. Accordingly, controller α allowed larger deviations in the knee and ankle kinematics compared to controller β .

IV. DISCUSSION

In this work, we derived an alternative parameterization of the impedance controller that more clearly illustrates its composite feedback and feedforward nature. By decomposing the impedance controller into these explicit components, we provided a unifying perspective that contextualizes impedance controllers along a spectrum between low-impedance torque controllers and high-impedance position controllers. From this perspective, we derived several key properties about impedance controllers that help to explain their behaviors and the particular roles of the impedance parameters, as demonstrated experimentally with an individual with an above-knee amputation walking on the Open-Source Leg v2.

These properties have important implications for the design, tuning, and understanding of impedance controllers for robotic prostheses. Below, we first explore how our alternative interpretation can improve controller design and tuning processes. We examine the challenge of selecting appropriate impedance parameters when multiple stiffness and damping solutions can produce equivalent nominal behaviors, and we suggest how online user-individualization methods could be improved. We then reconcile our experimental findings with the theoretical results, providing some practical insights and limitations to the theoretical findings. Finally, we discuss the broader implications of our framework for non-impedance-based control paradigms and its similarities to theories in human motor control.

A. Advantages of the Decoupled Parameterization

The decoupled form of the impedance controller in (3) offers advantages for controller design and tuning compared to the traditional spring-damper interpretation. Consider again

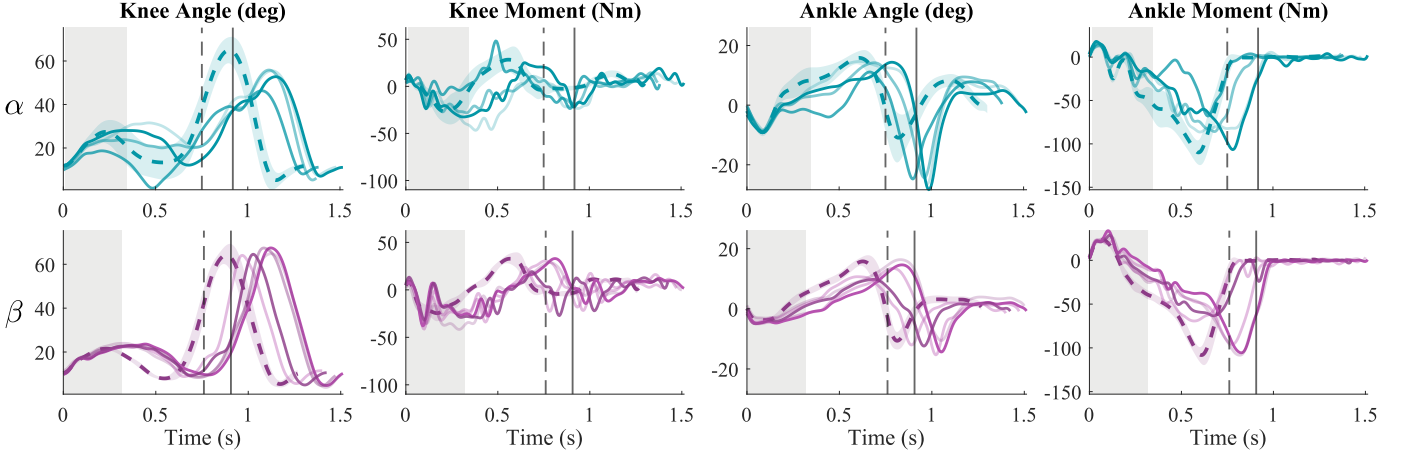


Fig. 7. The joint kinematic and kinetic traces for controller α (top row) and the controller β (bottom row) during perturbed and nominal walking. The dashed traces indicate the nominal, unperturbed trajectories for each condition, with the shading indicating ± 1 standard deviation. The solid lines denote trajectories from the perturbed strides, with corresponding trajectories from the same stride within each condition (across rows) shown in matching opacity. The gray shaded region indicates the average perturbation window (*i.e.* the time when the commanded treadmill speed deviated from nominal). The vertical solid and dashed lines indicate the average toe-off time for the perturbed and nominal conditions, respectively.

the tuning scenario where a researcher determines that the ankle torque should be increased at a specific point in the gait cycle to improve performance. While increasing the stiffness K or modifying the equilibrium angle θ_{eq} in (1) may seem like equivalent approaches, these modifications actually result in fundamentally different controller behaviors.

If the designer increases K while keeping θ_{eq} constant, they are simultaneously modifying both the feedback behavior and the feedforward torque component (via the $K\hat{\theta}$ term in τ_{ff}). This coupling makes it difficult to achieve targeted modifications. Our alternative parameterization eliminates this coupling by explicitly separating the feedforward component τ_{ff} from the feedback gains K and B . The designer can directly modify τ_{ff} without affecting the controller's response to perturbations. Conversely, if they want to make the controller more responsive to kinematic deviations, they can adjust K and B independently of the nominal behavior. This separation simplifies the tuning process and provides more precise control over the controller's characteristics.

The advantages of this decoupled parameterization extend beyond manual tuning to data-driven controller design approaches. When fitting impedance parameters using optimization methods, the traditional form (1) creates a non-convex constraint due to the product of K and θ_{eq} . This non-convexity can lead to local minima and makes global optimization challenging. However, using our alternative form (3) with τ_{ff} as a parameter instead of θ_{eq} results in an affine constraint, transforming the parameter fitting problem into a convex quadratic program that guarantees global optimality [9], [26]. This convex formulation has enabled the design of effective data-driven impedance controllers for variable speed/incline walking [9], sitting/standing [13], and stair climbing [14].

B. Non-Uniqueness During Nominal Conditions

1) *Multiple Impedance Solutions for a Nominal Behavior:* Another implication of this work is that the impedance parameters for a given nominal torque and angle behavior

are inherently non-unique. As demonstrated in Corollary 3.1, multiple sets of stiffness and damping parameters can produce identical nominal behaviors, making the selection of appropriate impedance values an under-constrained problem if we only consider these nominal behaviors. Given this non-uniqueness, how should the stiffness and damping be chosen?

Intuition suggests that stiffness and damping parameters that are either too high or too low are likely problematic for real-world use. Controllers with excessively low impedance magnitude may provide insufficient resistance to unexpected forces, allowing them to easily perturb the prosthesis's joint away from desired kinematics. For example, if a user with a low-impedance ankle controller encounters an unexpected obstacle, the controller may not provide adequate restoring forces to maintain stable locomotion. Conversely, controllers with excessively high impedance may respond too aggressively to minor kinematic variations, creating rigid and non-biomimetic behaviors that could be destabilizing or uncomfortable for the user. For instance, an ankle controller with very high stiffness might resist necessary joint motion when walking on uneven surfaces (*e.g.*, cobblestone streets or outdoor terrain), leading to harsh ground interactions rather than compliant adaptation.

Determining the optimal stiffness and damping values between these extremes remains an open question. The key insight from our analysis is that steady-state considerations alone are insufficient to guide this selection process due to the inherent non-uniqueness of the impedance parameters. Instead, stiffness and damping should be chosen to achieve desired feedback responses that help attenuate perturbations and return the user to nominal conditions. What those feedback responses should be remains an exciting avenue for future research.

2) Implications for Controller Evaluation and Tuning:

Given this observation, traditional evaluation and tuning approaches that rely only on steady-state performance metrics such as walking symmetry, metabolic cost, or kinematic similarity to able-bodied norms may miss critical differences in behavior produced by two different impedance controllers. To

truly compare controller behaviors with different stiffness and damping selections, evaluation protocols should include explicit gait perturbations and behaviors that drive the kinematics to depart from nominal. While users naturally exhibit some kinematic variation during nominal walking, this variation may be insufficient to reveal meaningful differences between impedance parameter choices. In our experimental validation, for example, the user's natural stride-to-stride variation did not elicit major differences between parameter sets during nominal conditions (Fig. 6) compared to during the perturbations (Fig. 7). Such targeted evaluation protocols might examine responses to external perturbations like pushes or trips, environmental variations like surface irregularities, or dynamics disturbances, like carrying loads. Even the simple treadmill speed perturbations used in this study were sufficient to elucidate behavior differences between stiffness and damping conditions, making this a simple and accessible option for most gait laboratories equipped with programmable treadmills.

The insights from our analysis also reveal potential limitations in existing impedance controller individualization and fine-tuning approaches. A significant body of literature focuses on adjusting stiffness and damping parameters while users walk under nominal conditions, using steady-state measures to guide parameter selection [1], [7], [27]–[32]. While these methods can effectively optimize the feedforward torque behavior (and thus the nominal walking pattern), they may miss the primary functional effects of the stiffness and damping parameters by not evaluating the system's response to perturbations. Since stiffness and damping can, in theory, have no effect on the nominal behavior (Property 3), there is essentially no gradient to guide the selection of K and B without the introduction of perturbations. This limitation could lead to the selection of impedance parameters that appear optimal during controlled laboratory tuning sessions but perform poorly in real-world scenarios where perturbations and disturbances are inevitable.

Future automated tuning frameworks may benefit from incorporating the decoupled parameterization. Rather than treating impedance tuning as a single optimization problem, the decoupled form suggests a hierarchical approach. First, the feedforward torque component τ_{ff} could be optimized to achieve desired nominal behaviors using conventional metrics like metabolic cost or kinematic similarity to able-bodied walking. Subsequently, the feedback gains K and B could be tuned specifically for perturbation rejection and stability around the nominal kinematics using controlled perturbation protocols. This approach would ensure that both the nominal performance and robustness characteristics are explicitly addressed in the tuning process.

C. Reconciling Theoretical and Experimental Results

Our experiments with a knee-ankle prosthesis user largely confirmed the theoretical claims made in Section II. The two controllers (α and β) produced very similar steady-state joint kinematics and kinetics (Fig. 6) despite having drastically different impedance parameters (Fig. 3). This result confirms that, given appropriate feedforward torque commands τ_{ff} to

produce the desired kinematics, the selected feedback behavior of K and B has little effect (Corollary 3.1).

The markedly different perturbation responses of the two controllers (Fig. 7) further supported our theoretical claims. These differences aligned with expectations based on the controllers' impedance parameters: controller α had lower stiffness during early stance and during toe-off, resulting in less aggressive position regulation and larger kinematic deviations from the nominal trajectories. The higher-impedance controller β behaved more like a PD position controller, maintaining the kinematic trajectories closer to their unperturbed counterparts aside from phase shifts. These observations demonstrate that perturbations are indeed necessary to reveal the functional differences between controllers with equivalent nominal behaviors.

However, our experimental results also revealed subtle differences in each controller's behavior even at steady-state. For example, we observed that controller α exhibited higher kinematic and kinetic variability compared to controller β (Fig. 6). Controller α 's lower impedance likely allowed more kinematic variation in response to the user's natural stride-to-stride variability. As the user/prosthesis system is highly non-linear, small kinematic changes can lead to larger subsequent kinematic and kinetic differences. Additionally, we did not know the exact feedforward torque τ^* required to produce the desired kinematics for our participant walking on the OSL. The able-bodied dataset used to estimate τ^* did not account for the dynamics of the OSL, nor amputee gait compensations. Thus, some kinematic deviation was inevitable, causing the different feedback behaviors to emerge in practice. These factors are likely true of any prosthesis used in the real world, meaning that stiffness and damping will inevitably have some impact on steady-state walking. However, this impact is relatively small compared to their drastic differences during perturbations.

D. Broader Implications Beyond Impedance Control

1) *Applications to Non-Impedance-Based Controllers:* Our results also have implications for prosthesis control paradigms beyond traditional impedance controllers, including ways to make emergent data-driven and model-free controllers (e.g., [33]) more robust for real-world use. Based on Property 2, these model-free control policies also inherently define joint impedance trajectories about some kinematic path. For the example in Fig. 2a-b, we saw that with torque matching as the only training objective, our neural network converged to a behavior with negative stiffness and damping at certain points in the gait cycle for our nominal walking kinematics. While the policy fit the training torque data well, this negative impedance could be problematic in online use.

First, users rarely interact with systems with negative impedance,⁴ potentially making acclimation more difficult with this policy. Further, a combination of Properties 1 and 2 shows that, locally, this negative impedance acts as an unstable position controller in (3) that pushes the kinematics away from

⁴While true negative impedance is impossible in passive physical systems, it is possible to create such systems, within some energetic bounds, with closed-loop control.

the desired trajectory when any deviation occurs. It is likely that a similarly-performing policy with positive impedance could be found by encoding desirable impedance properties in the loss function, which may be advantageous in producing more robust and intuitive online behavior.

More generally, our framework suggests that all joint-space prosthesis controllers exist somewhere on the impedance magnitude spectrum, with feedforward torque controllers on one extreme with $\|Z\| = 0$ and high-gain position controllers on the other extreme with $\|Z\| \gg 0$ (Fig. 4). Understanding where a particular controller lies on this spectrum can provide valuable insights into its expected behavior under perturbations, even if the controller was not explicitly designed using impedance concepts. Considerations should be prioritized to accommodate inevitable kinematic perturbations during real-world use, whether caused by external forces or by discrepancies in the system dynamics.

2) *Similarities with Human Motor Control*: Finally, it is interesting to note the similarities between the impedance controller parameterizations discussed in this work and well-studied aspects of human physiology and motor control. First, skeletal muscle-tendon units have long been shown to have spring- and damper-like responses to displacements, such as the linear “short-range stiffness” that occurs at small displacements [34], [35]. Joint-level perturbation studies during gait have produced similar findings, where observed changes in torque can be well explained by an impedance control law [18], [21], [22]. The ability to emulate these biological traits may be one of the reasons that impedance controllers have been successful for robotic prostheses.

Further, the two forms of the impedance controller discussed in this work—the traditional form parameterized by K, B, θ_{eq} and the proposed decoupled form parameterized by $K, B, \tau_{ff}, \theta_d, \dot{\theta}_d$ —are consistent with two prominent theories in human motor control. The traditional parameterization of the impedance controller is similar to the equilibrium point hypothesis, where the motor control system regulates muscle recruitment thresholds to generate desired mechanical behaviors [36], [37]. The resulting joint torque and equilibrium position arise from interactions with the environment, rather than from explicit control calculations. In contrast, the decoupled form of the impedance controller is more consistent with the internal model theory [38], in which motor commands similar to our τ_{ff} are directly calculated using an internal dynamic model to achieve a desired task (*e.g.*, [39]). Indeed, there is substantial overlap between human motor control theories and the control strategies for robotic leg prostheses, and advancements in either field may yield valuable insights in the other.

V. CONCLUSION

The novel parameterization of the impedance controller presented in this article provides a useful decoupling of its feedback and feedforward behaviors. This form helps contextualize impedance controllers between torque and position control paradigms, showing that impedance controllers can provide a first-order approximation to a generic torque control policy, and that position controllers are a subset of impedance

controllers where $\tau_{ff} = 0$. During nominal conditions with an appropriately selected τ_{ff} , the impedance controller reduces to a feedforward torque controller and the K and B parameters have little effect on the resulting behavior, both in theory and in practice. However, when the joint kinematics deviate from nominal trajectories, differences in the K and B parameters can result in large changes in behavior. Designing and tuning impedance controllers using our proposed parameterization rather than the standard K, B , and θ_{eq} parameterization may result in better outcomes because the feedback and feedforward behaviors can be individually modified. Further, methods that select K and B purely via observations of nominal behaviors may be insufficient, as the primary effects of changing these parameters are only visible during perturbations.

ACKNOWLEDGMENTS

The authors thank Dr. Curt Laubscher and Dr. Laksh Punith for feedback on our preliminary drafts of this article, and Ellie Wilson and Varun Shetty for assistance with the experiments.

REFERENCES

- [1] R. Gehlhar, M. Tucker, A. J. Young, and A. D. Ames, “A review of current state-of-the-art control methods for lower-limb powered prostheses,” *Annu. Rev. Control.*, vol. 55, pp. 142–164, 2023.
- [2] I. Fagioli, A. Mazzarini, C. Livolsi, E. Gruppioni, N. Vitiello, S. Crea, and E. Trigili, “Advancements and Challenges in the Development of Robotic Lower Limb Prostheses: a Systematic Review,” *IEEE Trans. Medical Robotics and Bionics*, vol. 6, no. 4, pp. 1409–1422, 2024.
- [3] Á. Einarsson and D. Langlois, “Methods and systems for controlling a prosthetic or orthotic device,” U.S. Patent Application US20220378588A1, 2022.
- [4] N. Hogan, “Impedance Control: An Approach to Manipulation,” in *American Control Conference*. IEEE, 1984, pp. 304–313.
- [5] F. Sup, A. Bohara, and M. Goldfarb, “Design and control of a powered transfemoral prosthesis,” *Int. J. Robotics Research*, vol. 27, no. 2, pp. 263–273, 2008.
- [6] S. Au, M. Berniker, and H. Herr, “Powered ankle-foot prosthesis to assist level-ground and stair-descent gaits,” *Neural Networks*, vol. 21, no. 4, pp. 654–666, 2008.
- [7] A. M. Simon, K. A. Ingraham, N. P. Fey, S. B. Finucane, R. D. Lipschutz, A. J. Young, and L. J. Hargrove, “Configuring a powered knee and ankle prosthesis for transfemoral amputees within five specific ambulation modes,” *PLoS ONE*, vol. 9, no. 6, p. e99387, 2014.
- [8] N. Anil Kumar, S. Patrick, W. Hong, and P. Hur, “Control framework for sloped walking with a powered transfemoral prosthesis,” *Front. Neurobot.*, vol. 15, no. Jan, pp. 1–12, 2022.
- [9] T. K. Best, C. G. Welker, E. J. Rouse, and R. D. Gregg, “Data-driven variable impedance control of a powered knee-ankle prosthesis for adaptive speed and incline walking,” *IEEE Trans. Robot.*, vol. 39, no. 3, pp. 2151–2169, 2023.
- [10] R. R. Posh, J. A. Tittle, D. J. Kelly, J. P. Schmiedeler, and P. M. Wensing, “Hybrid volitional control of a robotic transtibial prosthesis using a phase variable impedance controller,” in *IEEE Int. Conf. on Robotics and Automation*, 2024, pp. 4555–4561.
- [11] E. R. Westervelt, J. W. Grizzle, C. Chevallereau, J. H. Choi, and B. Morris, *Feedback Control of Dynamic Bipedal Robot Locomotion*. CRC Press, 2007.
- [12] D. J. Villarreal, H. A. Poonawala, and R. D. Gregg, “A Robust Parameterization of Human Gait Patterns Across Phase-Shifting Perturbations,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 3, pp. 265–278, 2017.
- [13] C. G. Welker, T. K. Best, R. D. Gregg, and C. G. Welker, “Improving Sit/Stand Loading Symmetry and Timing Through Unified Variable Impedance Control of a Powered Knee-Ankle Prosthesis,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 4146–4155, 2023.
- [14] R. J. Cortino, T. K. Best, and R. D. Gregg, “Data-driven phase-based control of a powered knee-ankle prosthesis for variable-incline stair ascent and descent,” *IEEE Trans. Med. Robot. Bionics*, vol. 6, no. 1, pp. 175–188, 2024.

- [15] E. Reznick, K. R. Embry, R. Neuman, E. Bolívar-Nieto, N. P. Fey, and R. D. Gregg, "Lower-limb kinematics and kinetics during continuously varying human locomotion," *Sci. Data*, vol. 8, 2021.
- [16] J. Camargo, A. Ramanathan, W. Flanagan, and A. Young, "A comprehensive, open-source dataset of lower limb biomechanics in multiple conditions of stairs, ramps, and level-ground ambulation and transitions," *J. Biomech.*, vol. 119, p. 110320, 2021.
- [17] E. J. Rouse, R. D. Gregg, L. J. Hargrove, and J. W. Sensinger, "The difference between stiffness and quasi-stiffness in the context of biomechanical modeling," *IEEE Trans. Biomedical Engineering*, vol. 60, no. 2, pp. 562–568, 2013.
- [18] H. Lee, E. J. Rouse, and H. I. Krebs, "Summary of Human Ankle Mechanical Impedance during Walking," *IEEE J. Trans. Eng. in Health and Med.*, vol. 4, pp. 1–7, 2016.
- [19] T. K. Best, G. C. Thomas, S. R. Ayyappan, R. D. Gregg, and E. J. Rouse, "A Compensated Open-Loop Impedance Controller Evaluated on the Second-Generation Open-Source Leg Prosthesis," *IEEE ASME Trans. Mechatron.*, pp. 1–13, 2024.
- [20] Y. Nazon, "Estimation of knee joint impedance during walking and its implications for robotic control and beyond," Ph.D. dissertation, University of Michigan, Ann Arbor, Michigan, 2024.
- [21] E. J. Rouse, L. J. Hargrove, E. J. Perreault, and T. A. Kuiken, "Estimation of human ankle impedance during the stance phase of walking," *IEEE Trans. Neural Sys. and Rehabil. Eng.*, vol. 22, pp. 870–878, 2014.
- [22] A. L. Shorter and E. J. Rouse, "Mechanical impedance of the ankle during the terminal stance phase of walking," *IEEE Trans. on Neural Sys. and Rehabil. Eng.*, vol. 26, no. 1, pp. 135–143, 2018.
- [23] H. Lee and N. Hogan, "Time-varying ankle mechanical impedance during human locomotion," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 23, no. 5, pp. 755–764, 2015.
- [24] V. Raveendranathan, V. G. M. Kooiman, and R. Carloni, "Musculoskeletal model of osseointegrated transfemoral amputees in OpenSim," *PLOS ONE*, vol. 18, no. 9, p. e0288864, 2023.
- [25] J. Camargo, K. Bhakta, J. Maldonado-Contreras, S. Zhou, K. Herrin, and A. Young, "OpenSim Model for Biomechanical Analysis with the Open-Source Bionic Leg," in *2022 Int. Symp. on Med. Rob.*, 2022, pp. 1–6.
- [26] N. Aghasadeghi, H. Zhao, L. J. Hargrove, A. D. Ames, E. J. Perreault, and T. Bretl, "Learning impedance controller parameters for lower-limb prostheses," *IEEE Int. Conf. Intelli. Rob. Sys.*, pp. 4268–4274, 2013.
- [27] H. Huang, D. L. Crouch, M. Liu, G. S. Sawicki, and D. Wang, "A Cyber Expert System for Auto-Tuning Powered Prosthesis Impedance Control Parameters," *Ann. Biomed. Eng.*, vol. 44, no. 5, pp. 1613–1624, 2016.
- [28] R. Wu, M. Li, Z. Yao, W. Liu, J. Si, and H. Huang, "Reinforcement learning impedance control of a robotic prosthesis to coordinate with human intact knee motion," *IEEE Robotics and Automation Letters*, vol. 7, no. 3, pp. 7014–7020, 2022.
- [29] M. Jacobson, P. Kantharaju, H. Jeong, J.-K. Ryu, J.-J. Park, H.-J. Chung, and M. Kim, "Foot contact forces can be used to personalize a wearable robot during human walking," *Sci. Reports*, vol. 12, no. 1, p. 10947, 2022.
- [30] A. Alili, V. Nalam, M. Li, M. Liu, J. Feng, J. Si, and H. Huang, "A novel framework to facilitate user preferred tuning for a robotic knee prosthesis," *IEEE Trans. Neural Sys. and Rehabil. Eng.*, vol. 31, pp. 895–903, 2023.
- [31] E. Reznick, T. K. Best, and R. D. Gregg, "A clinical tuning framework for continuous kinematic and impedance control of a powered knee-ankle prosthesis," *IEEE J. of Trans. Eng. in Health and Med.*, vol. 13, pp. 227–236, 2025.
- [32] W. Hong, A. Alili, and H. H. Huang, "An offline reinforcement learning-based auto-tuning framework for continuous impedance control in powered prostheses," in *Int. Conf. Rehabili. Rob.*, 2025, pp. 1829–1834.
- [33] C. P. O. Nuesslein and A. J. Young, "A deep learning framework for end-to-end control of powered prostheses," *IEEE Robotics and Automation Letters*, vol. 9, no. 5, pp. 3988–3994, 2024.
- [34] P. M. H. Rack and D. R. Westbury, "The short range stiffness of active mammalian muscle and its effect on mechanical properties," *The Journal of Physiology*, vol. 240, no. 2, pp. 331–350, 1974.
- [35] F. E. Zajac, "Muscle and tendon: properties, models, scaling, and application to biomechanics and motor control," *Critical Reviews in Biomedical Engineering*, vol. 17, no. 4, pp. 359–411, Dec. 1989.
- [36] A. G. Feldman and M. L. Latash, "Testing hypotheses and the advancement of science: recent attempts to falsify the equilibrium point hypothesis," *Experimental Brain Research*, vol. 161, no. 1, pp. 91–103, 2005.
- [37] A. G. Feldman and M. F. Levin, *The Equilibrium-Point Hypothesis – Past, Present and Future*. Boston, MA: Springer US, 2009, pp. 699–726.
- [38] D. M. Wolpert, Z. Ghahramani, and M. I. Jordan, "An internal model for sensorimotor integration," *Science*, vol. 269, no. 5232, pp. 1880–1882, 1995.
- [39] E. Todorov, "Optimality principles in sensorimotor control," *Nature Neuroscience*, vol. 7, pp. 907–915, 2004.